

Course Number	0418-426	Course Title	Embedded Microprocessor Systems
Credit Hours	3 (3:0:3)	Course Coordinator	Dr. Naser AlDuaij
Compulsory	No	Prerequisite	0418-325

Course Description

The subject area of this course is microprocessor systems and their embedding into technical equipment. The course continues and deepens subjects, which were covered in the Computer Systems and Programming courses, and concentrates on concepts to put these subjects together in order to design technical systems controlled by microprocessors. The course starts with the review of microprocessors, their internal structure, the instruction set, and continues with the memory and bus system, the I/O concept, the interrupt system, and controllers to interface them to peripheral devices. The course includes the related design, programming and test concepts for this area of application of microprocessor systems.

Textbook

Title: *Embedded Systems with ARM Cortex-M Microcontrollers in Assembly Language and C, Latest Edition*
Author: Y. Zhu
Publisher: E-Man Press LLC

Reference

Title: *Introduction to Embedded Systems, A Cyber-Physical Systems Approach, Latest Edition*
Author: E. Lee, S. Seshia
Publisher: MIT Press

Major Topics Covered in the Course

	Topics	Hours
1	Introduction to embedded systems: Purpose (IoT, smart home, wearables, robots, etc...), types of processors, microcontrollers and microprocessors (e.g., Arduino and Raspberry Pi), basics of running a program, data representation, ISAs	6
2	Programming embedded systems: Memory addressing, endianness, assembly instructions, registers and stack in assembly, flow control, arithmetic and logical instructions, subroutines, fixed-point arithmetic, floating-point arithmetic, instruction encoding/decoding	9
3	Programming embedded systems: Mixing C and assembly (e.g., ARM)	6

4	Hardware/software interfacing: Hardware and software interrupts, interrupt service routine, general-purpose I/O (GPIO), LEDs, LCDs, sensors and other embedded components, general-purpose timers: PWM, real-time clock, direct access memory (DMA)	9
5	Hardware/software interfacing: Analog I/O: Analog-to-digital (ADC) and digital-to-analog (DAC) converters, bus or serial communication protocols such as UART, I2C, SPI, USB	6
6	Embedded systems topics: Multitasking and concurrency, digital signal processing (DSP), embedded systems security	3
7	Exams	3
	Total	42

Grading

15-20%	Midterm 1
15-20%	Midterm 2
20-30%	Homework/Quizzes/Project
40%	Final

Course Outcomes

The student should be able

1. To analyze the functionality of complex devices and systems with embedded microprocessors.
2. To understand the concepts of implementing functions in hardware or in software.
3. To apply the knowledge and experience gathered from this course for the design, implementation and assessment of such complex.

Relationship between Course Outcomes and Student Outcomes

Course Outcomes	Student Outcomes					
	1	2	3	4	5	6
1	H					
2	L	H				
3		H				

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